

PSSE Contingency Analysis - Year 0 Topology 0

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Chapter 1

Introduction

1.1 Study Description

The Year 0 Topology 0 contingency analysis of the hypothetical SAVNW system is carried out to report the branches reporting loading greater than 100% and buses reporting upper and lower voltage limit violations.

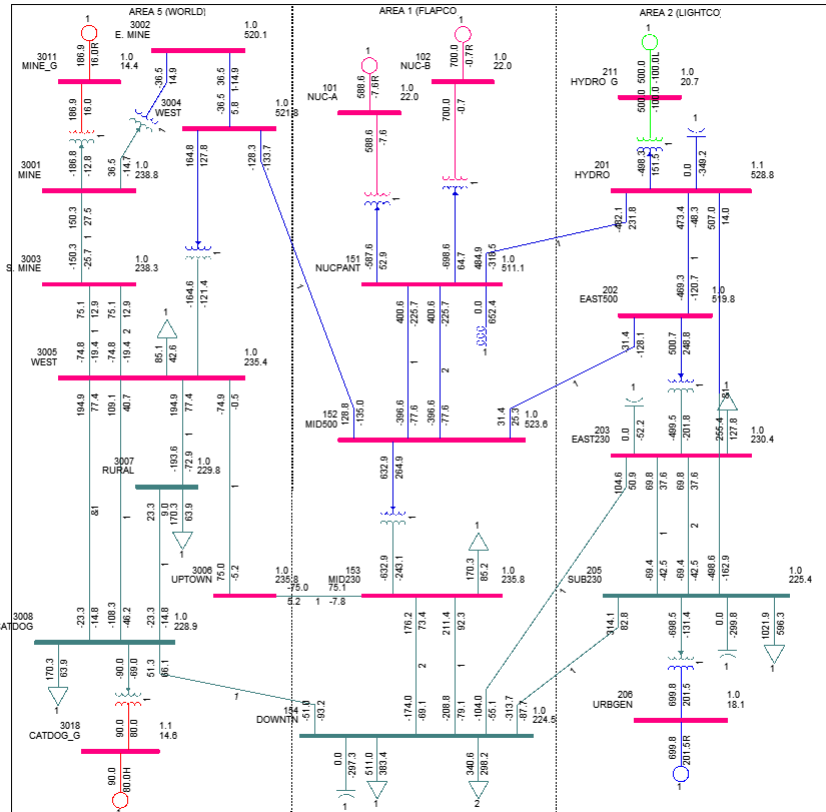


Figure 1.1: Single Line Diagram Year 0, Topology 0

For the hypothetical SAVNW system, the Year 0 Topology 0 of the system represents the present year system as shown in Figure 1.1. There are 6 generators in 3 areas. Total installed capacity of the system is the sum of firm capacity provided by 6 generating units and equals 4153.25 MW.

Detailed analysis of Year 0 Topology 0 base case was carried out and can be found here. The combined total generation in all areas of 2765 MW is serving a total load of 2723 MW with the generators 102 NUC-B, 211-HYDRO_G, 3018-CATDOG_G operating at their nominal power output of 700 MW, 500 MW and 90 MW respectively. The swing generator in each area operates to meet the load, loss and interchange criteria. The desired

interchange for all the scenarios studied is highlighted in Figure 1.2. For all the scenarios studied the desired interchange from FLAPCO to LIGHTCO is 100 MW, and from FLAPCO to WORLD is 150 MW.

X--	AREA	--X	FROM GENE- RATION	-----AT FROM IND GENERATN	AREA TO IND MOTORS	BUSES----- TO LOAD	TO BUS SHUNT	TO GNE BUS DEVICES	TO LINE SHUNT	TO XFRMR MAGNE- TIZING	FROM CHARGING	TO LOSSES	-NET INTERCHANGE- TO TIE LINES	TO TIES + LOADS	DESIRED NET INT
1	FLAPCO		1288.6	0.0	0.0	1021.9	0.0	0.0	0.0	0.0	0.0	15.3	251.5	1103.0	250.0
			-8.2	0.0	0.0	766.7	355.0	0.0	-0.0	-0.0	878.1	317.3	-569.2	112.3	
2	LIGHTCO		1199.8	0.0	0.0	1277.3	0.0	0.0	0.0	0.0	0.0	21.1	-98.6	-950.2	-100.0
			101.5	0.0	0.0	724.1	-701.1	0.0	-0.0	-0.0	651.0	396.2	333.3	-348.2	
5	WORLD		276.9	0.0	0.0	425.7	0.0	0.0	0.0	0.0	0.0	4.1	-152.9	-152.9	-150.0
			96.0	0.0	0.0	170.4	-0.0	0.0	-0.0	-0.0	366.0	55.8	235.9	235.9	
COLUMN			2765.3	0.0	0.0	2724.8	0.0	0.0	0.0	0.0	0.0	40.5	0.0	0.0	0.0
TOTALS			189.3	0.0	0.0	1661.2	-346.1	0.0	0.0	0.0	1895.2	769.3	0.0	0.0	

Figure 1.2: System Totals for Year - 0, Topology - 0

Chapter 2

Contingency Analysis

AC contingency calculation was conducted on the hypothetical SAVNW system for Year 0 Topology 0. The configuration files used are:

- Subsystem file savnw.sub - Studied subsystems of the studied case defined via the Subsystem Definition data file (Figure 2.1)
- Monitor file savnw.mon - Monitored Element Data File identifies the branches that are to be monitored for flow violations and the buses that are to be monitored for voltage violations (Figure 2.2)
- Contingency file savnw.con - Contingency cases that are to be tested are defined in the Contingency Definition data file (Figure 2.3)

```
SUBSYSTEM CON
AREA 1
AREA 2
AREA 5
END

SUBSYSTEM MON
AREA 1
AREA 2
AREA 5
END

END
```

Figure 2.1: The subsystem file corresponding to Year 1 Topology 1

```
MONITOR VOLTAGE RANGE ALL BUSES 0.9 1.1
MONITOR ALL BRANCHES
MONITOR TIES FROM AREA 1 TO AREAS 2 5
MONITOR TIES FROM AREA 2 TO AREA 5
END
END
```

Figure 2.2: The monitored file corresponding to Year 1 Topology 1

```
SINGLE BRANCH IN AREA 1
SINGLE BRANCH IN AREA 2
SINGLE BRANCH IN AREA 5
SINGLE BUS IN AREA 1
SINGLE BUS IN AREA 2
SINGLE BUS IN AREA 5
SINGLE MACHINE IN AREA 1
SINGLE MACHINE IN AREA 2
SINGLE MACHINE IN AREA 5
END
```

Figure 2.3: The contingency file corresponding to Year 1 Topology 1

The API DFAX_2 is used to construct distribution factor data files corresponding to the case .sav file, and the defined .sub, .mon, .con configuration files. By running the AC contingency calculation function ACCC_WITH_DSP_3, the contingency solution output .acc files is obtained.

Python code to conduct AC contingency calculation is:

```
import psspy
psspy.case(r"" savnw.sav "")
psspy.fdns([0,1,0,0,0,0,99,0])
psspy.dfax_2([1,1,0],r"" savnw.sub"",r"" savnw.mon"",r"" savnw.con"",r"" savnw.dfx"")
psspy.accc_with_dsp_3(0.1,[0,0,0,0,1,2,0,0,0,2,0],r"" CON"",r"" savnw.dfx"",r"" savnw.acc"", "",
    "", "")
```

Using the contingency solution output file savnw.acc, the results is exported as an excel file for further analysis. The results exported are ACCC Analysis Summary, Monitored Branch Flow (MVA), Monitored Bus Voltage.

Python code to export AC contingency solution output file as excel is:

```
import psspy
import pssexcel
pssexcel.accc('savnw.acc', ['s','b'], colabel='',stype='contingency', busmsm=0.5, sysmsm=5.0,
    rating='a', namesplit=False,xlsfile='out.xlsx', sheet='', overwritesheet=True,
    show=False, ratecon='b',
    baseflowvio=False,basevoltvio=False, flowlimit=100.0, flowchange=0.0, voltchange=0
    .0,swdrating='a',
    swdratecon='b',baseswdfio=False,basenodevoltvio=False,overloadreport=False)
```

Contingency analysis was carried out in PSSE for the studied case for Year 0 Topology 0. It was seen that the power flow solution did not converge for some of the tested contingencies. For the studied case, 42 contingencies were converged out of the 47 contingencies tested, the contingencies not converged are UNIT 206(1), SING OPN LIN 14 205-206(1), BUS 152, BUS 154, and BUS 201.

For the converged contingencies for the studied case, the result of the contingency analysis is analysed to check for branch overload ($> 100\%$) and out of range bus voltage violations (lower emergency limit $< 0.9PU$ and upper emergency limit $> 1.1PU$). It was seen that for some of the contingencies there were no branch overload or bus voltage violations reported. Rest of the contingencies violating branch overload and bus voltage emergency range violation are reported. Chapter 3 gives the observed branch flow violations, upper and lower emergency voltage violations for the studied case and the Chapter 4 summarises the results of all the analysis carried out on Year 0, Topology 0 of the hypothetical SAVNW system.

Chapter 3

Contingency Limit Violations

3.1 Introduction

In this chapter, for Year 0, Topology 0, the branches that are loaded more than 100% of their rating, and buses violating the upper and lower emergency ratings are reported.

3.2 Branch Overload Violation

For Year 0, Topology 0, out of the 34 branches in the system, 8 branches reporting a loading greater than 100% are given in Table 3.1.

Table 3.1: Loading Violations Year 0 Topology 0

Branch	Contingency	MVA Flow	AMP Flow	Rate	Loading
3001-3003(1)	SING OPN LIN 1 101-151(1)	532.205	515.011	300.000	171.670
153-154(1)	SING OPN LIN 11 202-203(1)	-349.967	359.164	350.000	102.618
3001-3003(1)	BUS 151	1050.651	1057.534	300.000	352.511
3001-3011(1)	BUS 151	-1648.816	1648.816	1560.000	105.693
3003-3005(1)	BUS 151	517.274	528.764	350.000	151.075
3003-3005(2)	BUS 151	517.274	528.764	350.000	151.075
3005-3007(1)	BUS 151	-353.065	378.495	350.000	108.141
153-154(1)	BUS 203	-348.675	357.543	350.000	102.155
153-154(1)	BUS 205	-310.402	352.007	350.000	100.573
154-203(1)	BUS 205	289.401	328.191	250.000	131.277
3001-3003(1)	UNIT 101(1)	532.205	515.011	300.000	171.670

For Year 0, Topology 0 the branches reporting a loading greater than 130% are the branches 3001-3003(1), 3003-3005(1), 3003-3005(2), 154-203(1). The branch 3001-3003(1) reported the violation for single line open contingency SING OPN LIN 1 101-151(1), for bus fault BUS 151, and for unit fault UNIT 101(1). The branches 3003-3005(1) and 3003-3005(2) reported the branch loading violation for the bus fault BUS 151, and the branch 154-203(1) reported a violation for the bus fault BUS 205. Further analysis would be required to assess the steady state and transient stability of the system to these faults as the loading violation is quite high for all these contingencies with the bus fault BUS 151 reporting a loading of as high as 353.5%.

3.3 Upper Emergency Bus Voltage Violation

For Year 0, Topology 0, there were no buses reporting an upper emergency voltage violation of greater than 1.1 PU.

3.4 Lower Emergency Bus Voltage Violation

For Year 0, Topology 0, the only bus reporting lower emergency voltage violation is the bus 154 for a bus fault at bus 205 labelled BUS 205 as can be seen from Table 3.2.

Table 3.2: Lower Emergency Voltage Violations Year 0 Topology 0

Bus Number	Contingency	Base Voltage	Contingency Voltage	Deviation	Range Violation
154	BUS 205	0.976	0.882	-0.094	-0.018

Chapter 4

Conculsion

The contingency analysis was carried out on the hypothetical SAVNW study system for the Year 0, Topology 0 to study the bus fault, single line open fault, unit fault contingencies. Out of all the studied contingencies, the contingencies for which the system did not converge are the contingencies UNIT 206(1), SING OPN LIN 14 205-206(1), BUS 152, BUS 154, and BUS 201.

For the converged contingency scenarios, Chapter 3 reported the branch overload, upper and lower limit voltage violations. From the branch overload table, Table 3.1 branches reporting loading greater than 130% were the branches 3001-3003(1), 3003-3005(1), 3003-3005(2), and 154-203(1). There were no upper limit voltage violations reported for any of the studied buses. Lower limit voltage violations were reported by the bus 154 for a bus fault BUS 205 for Year 0 Topology 0 (Table 3.2).

The branch reporting the greatest number of overloaded violations are the branches 3001-3003(1) and 153-154(1) with 3 violations each as can be seen from Table 4.1. The contingency causing the most number of branch overload violation is the bus fault contingency BUS 151 with 5 overload violation followed by the bus fault contingency BUS 205 with 2 violations as can be seen from Table 4.2.

Appendix

Table 4.1: Overload Count - Branch

Branch	Overload Counts
3001-3003(1)	3
153-154(1)	3
3001-3011(1)	1
3003-3005(1)	1
3003-3005(2)	1
3005-3007(1)	1
154-203(1)	1

Table 4.2: Overload Count - Contingency

Contingency	Overload Counts
BUS 151	5
BUS 205	2
SING OPN LIN 1 101-151(1)	1
SING OPN LIN 11 202-203(1)	1
BUS 203	1
UNIT 101(1)	1